Sample Problem 1: School Results Application

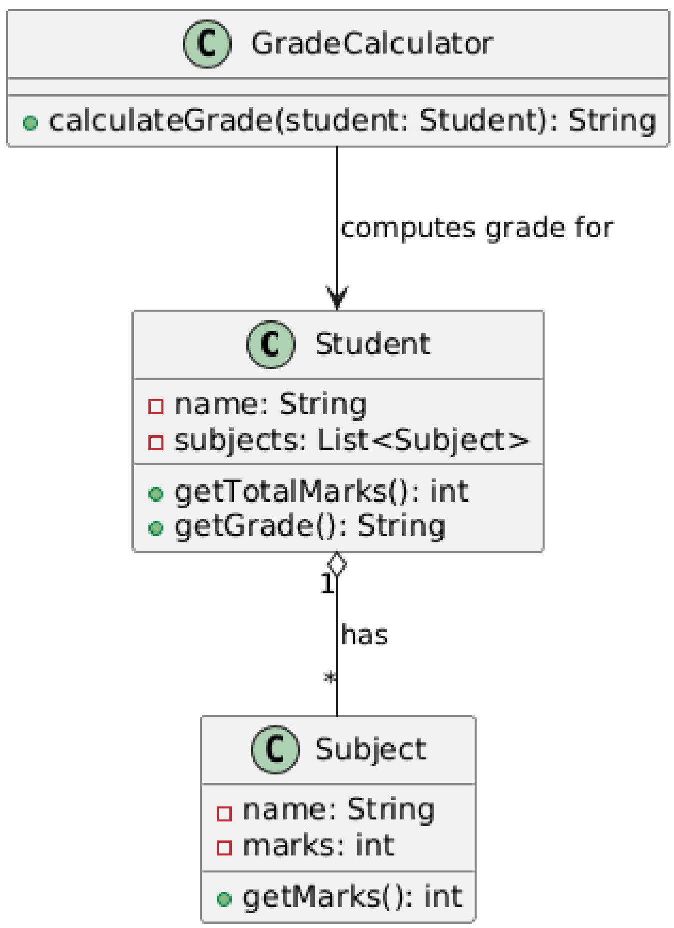
Class Diagram

The class diagram represents the structure of a school results application where students have subjects, and their scores are calculated for grades.

Diagram Description:

- **Classes**: Student, Subject, GradeCalculator
- **Relationships**:
  - A Student has multiple Subject entries (Aggregation).
  - GradeCalculator computes the results for a Student.

→ Class Diagram:



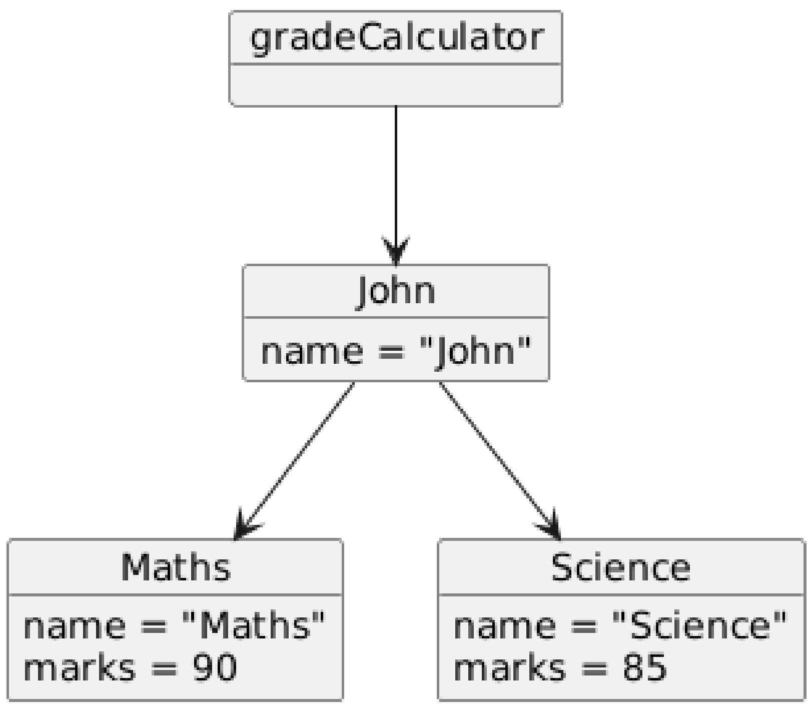
Object Diagram

An object diagram provides a snapshot of the Student and their Subject objects at a particular point.

Example:

- **Student**: John
- **Subjects**: Maths, Science
- **Marks**: 90, 85

→ Object Diagram:



Sequence Diagram

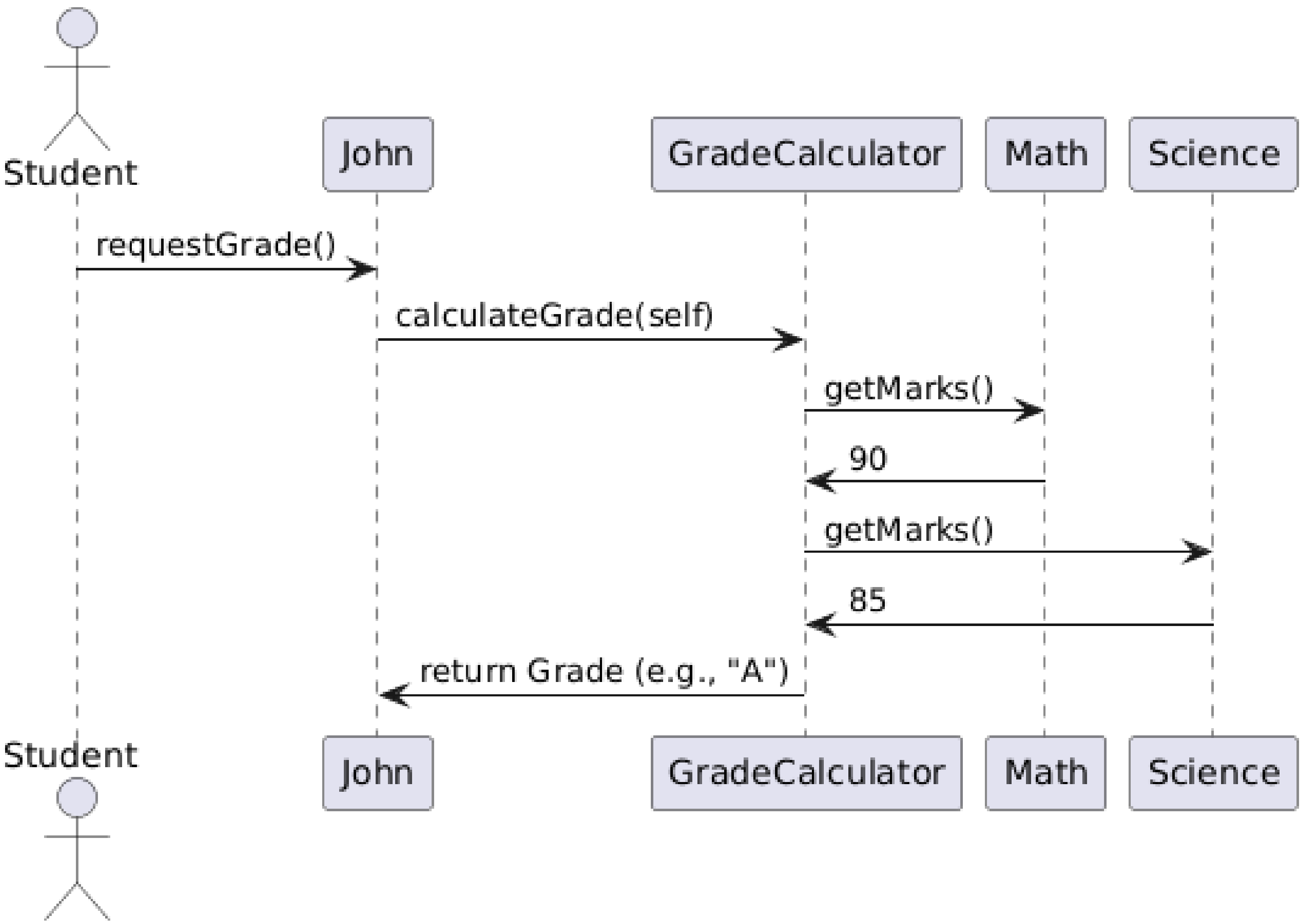
The sequence diagram shows how objects interact to calculate grades.

Scenario: A student requests their grade based on marks in subjects.

Actors:

1. Student
2. GradeCalculator

→ Sequence Diagram:



Sample Problem 2: Grocery Store Bill Generation Application

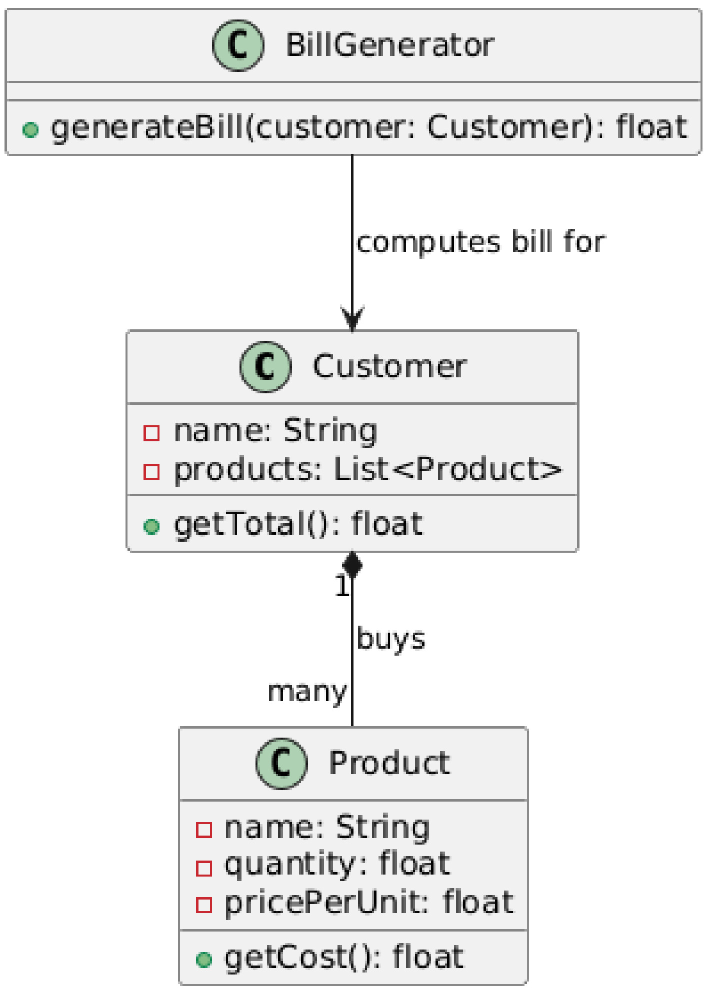
Class Diagram

The class diagram models the system where a customer buys products, and the bill is generated.

Diagram Description:

- **Classes**: Customer, Product, BillGenerator
- **Relationships**:
    - A Customer can purchase multiple Product items (Composition).
    - BillGenerator computes the total for the Customer.

→ Class Diagram:



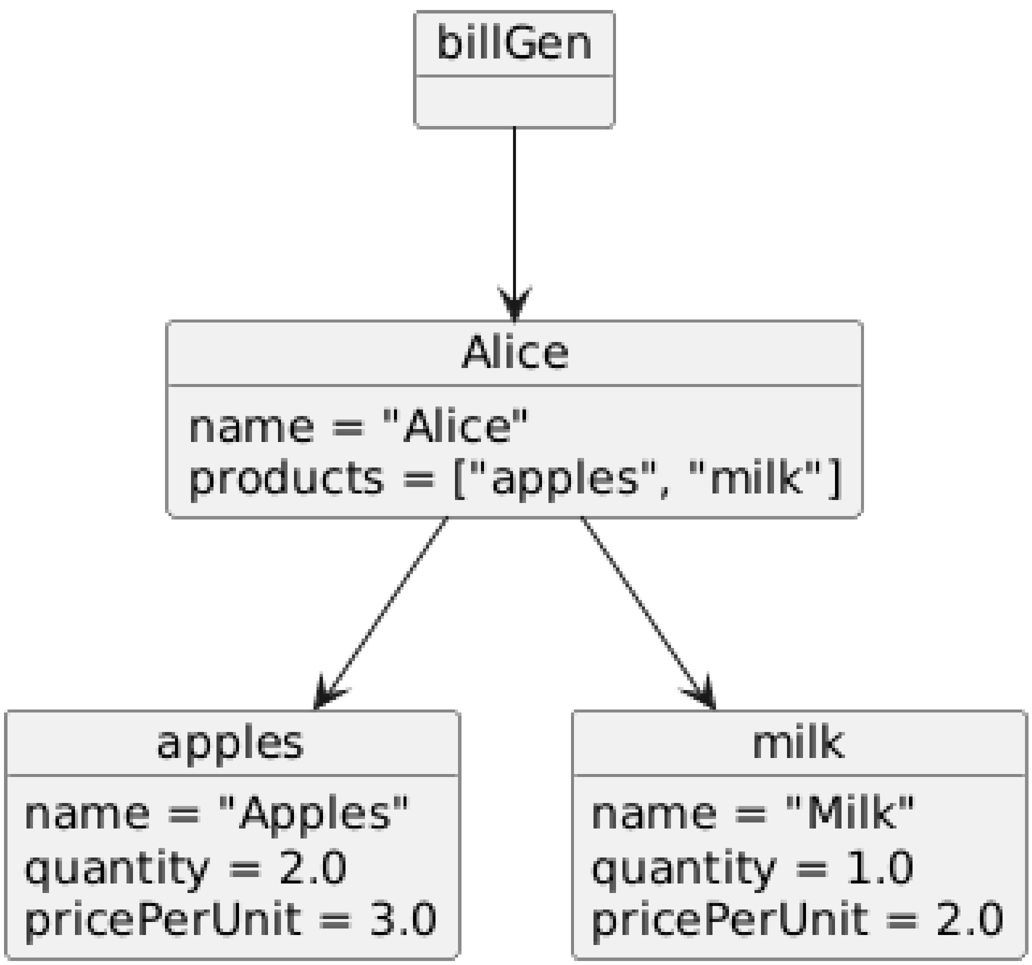
Object Diagram

An object diagram shows the details of a Customer and the Product objects they have purchased.

Example:

- **Customer**: Alice
- **Products**:
  - Apples (2 kg at $3 per kg)
  - Milk (1 liter at $2 per liter)

→ Object Diagram:



Sequence Diagram

The sequence diagram shows the process of bill generation for a customer.

Scenario: A customer checks out at the grocery store, and the total bill is generated.

Actors:

1. Customer
2. BillGenerator

→ Sequence Diagram